# Final Capstone Case

This task evaluates your ability to perform **DNS, URL, and network forensic analysis** using passive, legally compliant methods and to document findings in a professional manner.

---

## Case Materials (Provided Separately)

- PCAP file
- Alert files (text/image)
- Malware sample (**for hashing and identification only – do not execute**)

---

## Scenario

Network traffic was captured from a LAN segment belonging to the domain **phenomenoc[.]com**.

**LAN Details:**

- LAN range: 10.0.76.0/24
- Domain controller: 10.0.76.6 (Phenomenoc-DC)
- Gateway: 10.0.76.1

Suspicious activity was detected on the network, and an internal Windows host is believed to be infected.

---

## Investigation Objectives

Using the provided data, determine and document:

1. The IP address, MAC address, and host name of the infected Windows host
2. The Windows user account associated with the infected host
3. The malware delivery method
4. The external IP address used during malware delivery
5. The SHA256 hash of the retrieved executable
6. The malware type indicated by the alert data
7. The IP address involved in post-infection communication
8. The domain name used for post-infection traffic

---

## Required Tools

- Wireshark
- Zeek (optional, passive analysis only)
- NetworkMiner (open-source)
- Hashing utilities (e.g., sha256sum/VirusTotal)
- Text editor for analysis notes

---

## Analysis Constraints

- Passive analysis only
- No execution of malware samples
- No interaction with live systems
- No scanning, probing, or exploitation

---

## Deliverables

1. Evidence register (including hashes)
2. Analysis notes with supporting screenshots
3. List of identified Indicators of Compromise (IOCs)
4. Final technical investigation report
5. One-page non-technical summary

This case simulates a real incident response investigation. Your findings must be supported by evidence and clearly documented.